

Supply Chain Risk Report — Carbon Fiber Price Surge Contingency

Report ID: SCR-2026-0520-001 | Date: 2026-05-20
Product: Pokee ████████ (SKU-1) | Trigger: Toray T1100 Carbon Fiber +30% price increase

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1. Executive Summary

Toray Japan has announced a **30% price increase** on T1100 carbon fiber, the primary material for the main shaft of our flagship product **Pokee ████████ (SKU-1)**. This report quantifies the cost impact and outlines alternative supplier options to mitigate supply chain risk.

Critical Alert: A 30% material cost increase on the main shaft drives a 17.6% total product cost increase (\$25.50 per unit). Immediate action required to maintain margin targets.

2. Current BOM — SKU-1 (Pokee ████████)

Component	Material	Weight (g)	Unit Cost (USD)	Lead Time (days)	Supplier
Main Shaft	Toray T1100 Carbon Fiber	120	\$85.00	45	Toray Japan
Guides	Titanium Alloy + SIC	30	\$45.00	30	Fuji Guides
Handle	AAA Grade Cork	25	\$15.00	15	Portugal Cork Co.

Component	Material	Weight (g)	Unit Cost (USD)	Lead Time (days)	Supplier
Total		175	\$145.00		

3. Cost Impact Simulation (+30% Toray T1100)

Component	Original Cost	New Cost	Delta
Main Shaft (T1100)	\$85.00	\$110.50	+\$25.50
Guides	\$45.00	\$45.00	\$0.00
Handle	\$15.00	\$15.00	\$0.00
Total	\$145.00	\$170.50	+\$25.50 (+17.6%)

Key Finding: A 30% material cost increase on the main shaft drives a **17.6% total product cost increase** (\$25.50 per unit).

4. Alternative Supplier Recommendations

4.1 Carbon Fiber Substitutes for T1100

Alternative	Grade	Est. Unit Cost	Cost vs. T1100	Lead Time	Risk Level	Notes
Teijin T800 (Teijin Co., Japan)	High modulus	~\$78	-8.2%	40 days	Low	Mature supplier, slightly lower tensile strength (5,200 MPa vs 6,370 MPa)

Alternative	Grade	Est. Unit Cost	Cost vs. T1100	Lead Time	Risk Level	Notes
SGL T800S (SGL Carbon, Germany)	High performance	~\$82	-3.5%	35 days	Low	EU-based, shorter lead time, IATF 16949 certified
HexTow AS4S (Hexcel, USA)	Aerospace grade	~\$90	+5.9%	30 days	Medium	Premium pricing but stable long-term contracts available
Guangzhou Dayu CF (Dayu Composite, China)	Domestic high-grade	~\$55	-35.3%	20 days	High	Significant cost savings but requires full QA re-validation
Zhongwen Carbon (Zhongwen Co., China)	T700-class	~\$50	-41.2%	15 days	High	Lowest cost option, suitable for budget-tier product line

4.2 Recommended Strategy

Timeline	Action	Owner
Immediate (0-2 weeks)	Negotiate volume-based pricing with Toray Japan; lock current rates for 6-month supply	Procurement
Short-term (2-4 weeks)	Request samples from Teijin T800 and SGL T800S for performance testing	R&D; / QA
Medium-term (4-8 weeks)	Initiate qualification process for Dayu Composite as cost-reduction backup	Procurement
Long-term (8-12 weeks)	Establish dual-source supply chain: Toray (premium) + Chinese supplier (budget tier)	Supply Chain

5. Financial Impact Summary

Scenario	Unit Cost	Margin Impact	Annual Impact (est. 10,000 units)
Baseline (current)	\$145.00	—	—
+30% T1100 only	\$170.50	-25.5 pts	-\$255,000
Switch to Teijin T800	\$148.50	-3.5 pts	-\$35,000
Switch to Dayu CF	\$119.50	+25.5 pts savings	+\$255,000
Dual-source (60% Toray / 40% Dayu)	\$140.20	+4.8 pts	+\$48,000

6. Risk Matrix

Risk	Likelihood	Impact	Mitigation
Toray extends price hike beyond 30%	Medium	Critical	Dual-source strategy
Alternative supplier quality failure	Medium	High	Staged qualification, pilot batches
Lead time disruption (geopolitical)	Low	High	Safety stock 3-month buffer
Contract renegotiation delays	High	Medium	Parallel-track supplier onboarding

7. Approval & Next Steps

The following actions require sign-off from relevant stakeholders:

- Procurement to initiate Toray renegotiation by **2026-05-27**

- ■ R&D; to receive Teijin/SGL samples by **2026-06-10**
- ■ QA to complete alternative material testing by **2026-06-30**
- ■ Executive sign-off on dual-source strategy by **2026-07-15**

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